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# Evidence of vertical transmission and tissue tropism of Streptococcosis from naturally infected red tilapia (*Oreochromis* spp.)

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ABSTRACT

Streptococcosis is a highly problematic disease in the aquaculture of freshwater fishes, especially for tilapia. The possibility of vertical transmission of streptococcosis and the pattern of tissue tropism of this pathogen in various organs was examined in red tilapia (*Oreochromis* sp.). Healthy broodstock without any clinical signs of *Streptococcus* spp. were selected from a farm earlier reported to have the disease and a total of 10 pairs were forced spawned to provide samples of gametes and progeny for pathogen testing. A colorimetric LAMP assay was used to confirm whether the bacterial pathogens *Streptococcus. agalactiae* and *Streptococcus. iniae* was present in samples of milt, unfertilized eggs, fertilized eggs, and offspring at various stages of development, as well as internal organs of broodstock (reproductive organs, gill, liver, spleen, kidney and brain) as well as samples of water from culture systems. The majority of samples of milt (9/10) and unfertilized eggs (7/10) collected from the broodstock were infected with *S. iniae* at the time of spawning and was transmitted to all of their offspring. Nevertheless, when the same samples of gametes were analyzed for *S. agalactiae*, they were all found to be negative but the pathogen was found to be present in some 10-day-old larval offspring (4/10). However, when the pathogenic presence was analyzed from the reproductive organs of the parents, both *S. agalactiae* (11/20) and *S. iniae* (18/20) bacterium were common. Although, all broodstock were asymptomatic, almost all broodstock harboured the bacteria in many organs. Confirmation of vertical transmission of streptococcosis in tilapia means that intergenerational break cannot be used as a reliable and simple means of reducing or eliminating the prevalence of these difficult pathogens in aquaculture stock.



## 1. Introduction

Streptococcosis is a disease resulting from infection by members of a diverse group of bacteria (*Streptococcus* spp. and *Lactococcus* sp.) that possess the capacity to infect a wide range of hosts, including humans (Johri et al., 2006; Amal and Zamri-Saad, 2011; Plumb and Hanson, 2011). Streptococcosis outbreaks in tilapia have been reported from many parts of the world (Al-Harbi, 1994; Abuseliana et al., 2010; Eldar et al., 1994; Perera et al., 1994; Shoemaker et al., 2001) with the disease becoming a major problem in tilapia aquaculture with total global losses in production in 2008 estimated at US$250 M (Klesius et al., 2008; Shoemaker and Klesius, 1997). High mortality rates of more than 50–70% over a period of 3–7 days are common (Wongsathein, 2012; Yanong and Francis-Floyd, 2002)

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that are frequently associated with intensive aquaculture systems (Hernandez et al., 2009; Shoemaker et al., 2001; Salvador et al., 2005). The most significant causative bacteria of streptococcosis in fish worldwide includes; *S. iniae*, *S. agalactiae, S. parauberis, S. difficile*, *S. shiloi* and *S. dysgalactiae* (Abuseliana et al., 2010; Costa et al., 2014; Eldar et al., 1994; Evans et al., 2006; Figueiredo et al., 2012; Haines et al., 2013; Mata et al., 2004; Maisak et al., 2008; Netto et al., 2011; Zhang et al., 2003). In Thailand, where tilapia aquaculture is extensive, both *S. agalactiae* and *S. iniae* have been reported as the most prevalent causative species (Rodkhum et al., 2012; Suanyuk et al., 2008, 2010), however, they are difficult to distinguish due to their similar microbiological appearance and clinical signs. According to their biochemical characterization and properties, *S. agalactiae* belongs to Lancefield group B Streptococcus, Gram-positive cocci, with either β-or non-haemolytic (γ), non-motile, oxidase negative properties (Wongsathein, 2012), while, *S. iniae* is a non-Lancefield, Gram-positive coccus bacteria with catalase-and oxidase negative, non-motile and β-haemolytic properties (Agnew and Barnes, 2007). Clinical signs and symptoms of streptococcosis in fish may vary depending on the species of fish, however, the most common symptoms include loss of orientation and erratic swimming, unilateral or bilateral exophthalmia, anorexia, eye opacity, abdominal distention, darkening of skin and hemorrhaging skin around the anus or at the base of the fins (Amal and Zamri-Saad, 2011; Amal et al., 2013a; Evans et al., 2002; Karsidani et al., 2010).

The transmission of *Streptococcus* spp. in tilapia via a range of experimental routes has been demonstrated, including intraperitoneal injections, immersion, cohabitation with infected fish, as well as oral, gill and nare inoculation (Evans et al., 2000; Perera et al., 1994; Shoemaker et al., 2000). Under natural conditions the main pathways of disease transmission appears to be through direct contact between healthy fish and diseased or dead fish, as well as indirect contact via the water in culture systems (Agnew and Barnes, 2007,b; Amal et al., 2013a,b; Bowater et al., 2012; Kim et al., 2007; Nguyen et al., 2002; Robinson and Meyer, 1966). A number of studies confirm the horizontal transmission of *Streptococcus* spp. in tilapia (Amal et al., 2013b; Evans et al., 2000; Hernandez et al., 2009; Hossain et al., 2014; Jimenez et al., 2007; Mian et al., 2009; Shoemaker et al., 2000), whereas natural outbreaks of the disease on tilapia farms indicate that larvae and juvenile fish smaller than 20 g may not be susceptible (Hernandez et al., 2009; Jimenez et al., 2007), suggesting that little or no vertical transmission of the disease.

A recent study detected both *S. agalactiae* and *S. iniae* in the reproductive organs (testis and ovary) as well as in the fertilized eggs and fingerlings derived from apparently healthy broodstock of hybrids of Nile and red tilapia (Suebsing et al., 2013). These results suggest that vertical transmission of these bacterial diseases to progeny may also be possible in tilapia. To test this proposition, gametes were sampled from individual broodstock under quarantine conditions and used for *in vitro* fertilization and subsequent culture of offspring. Major organs of the broodstock as well as samples of milt, unfertilized eggs, fertilized eggs and progeny at various ages were screened for the presence of both *S. agalactiae* and *S. iniae* to determine the possible vertical transmission of these pathogens.

## 2. Materials and methods

### 2.1. Broodstock source and maintenance

A total of 40 female and 20 male adult red tilapia (*Oreochromis* sp.) of 150–200 g in size were randomly selected from a commercial farm stock held at Naam Sai Farm, in Prachinburi Province, in eastern Thailand. All of the adult fish were apparently healthy and had previously been routinely used as broodstock for the commercial production of fingerlings. They were then transferred in water to quarantine aquaculture research facilities where they were maintained in aerated 2 t round polyethylene tanks at a stocking density 12 $m^2$ with a sex ratio (male:female) of 1:3. The fish were conditioned for a month by feeding with standard commercial floating tilapia feed pellets (CP feeds) containing 28% crude protein at a rate of 4% of their biomass three times a day. In addition to the pelleted feed, fishes were also given duckweed (*Lemna* spp.) ad-libitum once a week as a supplementary natural feed (Pradeep et al., 2012). Water quality was maintained to ensure optimum conditions for the fish through adequate water exchange in the tanks salinity 0 ppt; total ammonia nitrogen <0.5 ppm; total nitrite <0.5 ppm; dissolved oxygen >5 ppm; pH 7.0–8.0. Filtered (1 μm) and UV sterilized bore water was used to minimize the potential for introduction of water-borne pathogens. Broodstock and larval rearing tanks for the experiment were strictly maintained under quarantine conditions in an isolated fish rearing experimental unit to reduce problems with internal contamination, where all the handling equipment, floors and tanks were routinely disinfected using iodine povidone as a bio-security measure.

### 2.2. Artificial fertilization

Synchronized breeding, gamete collection and artificial fertilization from the tilapia broodstock were followed according to protocols described previously (Pradeep et al., 2010) and were conducted under the quarantine conditions. The broodstock were starved for 24 h and the females were then inspected individually to determine the stage of maturity and only those individuals ready to spawn ($n = 10$) were selected for subsequent stripping of eggs into a dry sterilized Petri dish. For each female about 20 unfertilized eggs were then sub-sampled and transferred to a microcentrifuge tube (1.5 ml) for pathogen screening. A matching number of male fish ($n = 10$) were also selected and stripped of milt after blotting the fish dry to avoid any possible activation or contamination of the sperm, while withdrawing the milt directly into fine capillary tubes (6 tubes each of 50 μl) whilst taking extreme care to avoid any contamination with urine. Three capillary tubes of sperm were transferred to a microcentrifuge tube (1.5 ml) for pathogen screening. The remaining three capillary tubes of sperm from a male fish were then used to fertilize eggs from a female fish by spreading the sperm over the eggs placed in the Petri dish. To activate the sperm for artificially inseminating the eggs, 15 ml of UV sterilized fresh water (28 ± 1 °C) was added to the petri dish and was shaken gently for thorough mixing of the sperm with eggs. After two minutes, excess water was decanted from the petri dish and an additional 15 ml of UV sterilized fresh water (28 ± 1 °C) was added and then drained. After approximately 1 h 20 fertilized eggs were sub-sampled into a 1.5 ml microcentrifuge tube for subsequent pathogen screening, and the remaining eggs were carefully moved into a round-bottomed hatching chamber provided with UV sterilized water (Pradeep et al., 2011). Upon hatching, the larvae were moved from the hatching chamber to an individual larval rearing tank (100 l), which was maintained with filtered (1 μm) UV sterilized water and reared until the termination of the experiment after 30 days from hatch. The water quality in the larval culture tanks was closely monitored and any dead larvae or debris were removed at regular intervals. The larvae were feed with powdered feed (CP feeds) (28% protein) at a ration of 12–15% of their body weight three times a day for a period of 10 days and then the feeding ration was subsequently reduced to 8–10% of the total biomass.

### 2.3. Sample collection

A total of 10 pairs of tilapia broodstock were artificially bred and samples of gametes and larvae collected for testing for the

**Table 1**
Samples collected and analysed for presence of *S. agalactiae* and *S. iniae*. Paired broodstock (i.e., $n = 10$♀ and 10♂) and their gametes, offspring, blood and tissues were sampled for analysis.

| Sample | | Preparation for one determination | | n |
|---|---|---|---|---|
| | | Amount | Added into | |
| Unfertilized eggs | | 20 eggs (pooled) | 200 ml 50 mMNaOH | 10 |
| Fertilized eggs | | 20 eggs (pooled) | 200 ml 50 mMNaOH | 10 |
| Milt | | 50 ml | 200 ml 50 mMNaOH | 10 |
| 1-day-old larvae | | 15 (pooled) | 200 ml 50 mMNaOH | 10 |
| 10-day-old larvae | | 15 (pooled sample) | 200 ml 50 mMNaOH | 10 |
| 30-day-old larvae | | 5 (pooled sample) | 200 ml 50 mMNaOH | 10 |
| Rearing water | | | | |
| | From broodstock tank | 15 ml | | 10 |
| | From hatching chamber | 15 ml | | 10 |
| | From larval tank | 15 ml | | 10 |
| Broodstock | | | | |
| | Blood | 100 ml | 200 ml 50 mMNaOH | 20 |
| | Gills | 50 mg | 200 ml 50 mMNaOH | 20 |
| | Liver | 50 mg | 200 ml 50 mMNaOH | 20 |
| | Spleen | 50 mg | 200 ml 50 mMNaOH | 20 |
| | Anterior kidney | 50 mg | 200 ml 50 mMNaOH | 20 |
| | Brain | 50 mg | 200 ml 50 mMNaOH | 20 |
| | Ovary | 50 mg | 200 ml 50 mMNaOH | 10 |
| | Testis | 50 mg | 200 ml 50 mMNaOH | 10 |

presence of *S. agalactiae* and *S. iniae*. A 100 μl blood sample was also taken from the caudal vein of each parent using a 23-gauge needle into a 1 ml syringe (Nipro tuberculin syringe) containing 200 μl of 50 mMNaOH. One-day-old, 10-day-old larvae (approximately 15 individuals each) and 30-day-old larvae (5 individuals each) were collected from each of the 10 sets of larvae for pathogen screening (Table 1). In addition, 15 ml of rearing water was sampled from each of the five broodstock tanks (BT) on two occasions, and once each for the 10 hatching chambers (HC), and ten larval rearing tanks (ST) for pathogen screening. All the samples were stored at 4 °C until pathogen detection by LAMP assay. To determine the presence of infection of major organs with *S. agalactiae* and *S. iniae* in the 20 broodstock fish they were necropsied after cold euthanasia. Tissue (50 mg) from various organs including; gill, liver, spleen, anterior kidney, brain (medulla oblongata) and reproductive organs were sampled. All samples, excluding water samples, were collected in a micro-centrifuge tube (1.5 ml) and preserved in 200 μl of 50 mMNaOH and then stored at 4 °C.

### 2.4. DNA extraction and bacterial assay

DNA was extracted from all samples following a rapid extraction method using a 0.025 N sodium hydroxide and 0.0125% sodium dodecyl sulphate solution as previously reported (Jaroenram et al., 2009). For DNA extraction from water samples, the sample was first filtered through sterile 0.45 μmfilters (Millipore, Bedford, MA, USA) and then the filter paper was subjected to the rapid extraction method. Extracted nucleic acids were adjusted to a final concentration of 100 ng $\mu l^{-1}$ and were stored at −20 °C until use. The quality of the DNA was verified by polymerase chain reaction [cytochrome oxidase I gene (COI); Kowasupat et al., 2014]. For PCR amplifications and colorimetric loop-mediated isothermal amplification (LAMP) assay, 2 μl of the DNA template was used. Colorimetric LAMP assay for *S.agalactiae* and *S. iniae* detection were modified from the previously reported protocol (Suebsing et al., 2013). LAMP specific-primer sets for *S. agalactiae* (GenBank: HM004089–HM004094) and *S. iniae* ++ (GenBank: HM004083–HM00488) were designed according to the consensus sequences of the published sequence of the superseded disputes (*sod*A).

In brief, colorimetric LAMP assay was performed in a 25 μl of total reaction mixture containing 2 μM each of FIP and BIP, 0.2 μM each of F3 and B3, 2 μM each of LF and LB primers (Table 2), 1 × thermopol-supplied reaction buffer, 0.6 M betaine

**Table 2**
Oligonucleotide primers used for colorimetric LAMP for detection of *S. agalactiae* and *S.iniae.*

| Primer | Sequence (5'–3') |
|---|---|
| Sa-F3 | ATATGATGCGCTTGAGCC |
| Sa-B3 | ACCACCGTTATTGATGACTG |
| Sa-FIP | GAGCAGCATTTGCATTAGCAACATATTTTGATGCTGAGACAATGACAC |
| Sa-BIP | ACATCCTGAAATTGGAGAAGACTTTTTTCCTGACGAATATCTTCTGGAAT |
| Sa-LF | TGCATGGTGCTTATCATGATGT |
| Sa-LB | AGGCGCTCTTAGCTGATGT |
| Si-F3 | TAATGGTGGCGGACACTT |
| Si-B3 | AAGTAATTTCAAGACTTCCTTCT |
| Si-FIP | TGCACTTGCAACTTCTTTTGTTACTTTTTAATCATGCTTTGTTCTGGGA |
| Si-BIP | AAGAACAATTTGCAGCAGCAGCTTTTAACAACTAACCAAGCCCA |
| Si-LF | ACTGGCCGTTTTGGTTCTGGT |
| Si-LB | GACCAAGCTTTTGGATC |

(USB Corporation, Cleveland, Ohio, USA), 6 mM $MgSO_4$ (New England Biolabs, Ipswich, USA), 1.2 mM dNTPs mix (Thermo Fisher Scientific, Waltham, MA USA), 8 U Bst DNA polymerase (New England Biolabs, Ipswich, MA USA) and 125 μM hydroxynaphthol blue dye, Sigma–Aldrich, USA. A reaction mixture without template was included as negative control and 100 ng of DNA extracted from *S. agalactiae* (DMST 17129) and *S. iniae* (ATCC 29178) were included as positive controls to confirm the reliability of the assay. The mixtures were incubated at 63° C for 30 min, after which the colour of the reaction mixture could be detected visually. The color change from purple to blue was interpreted to be a positive sample (i.e., presence of the bacteria), while samples which remained purple were interpreted as negative. For comparison, the LAMP amplicons were detected by the UV–visible spectral analysis (Thermo Fisher Scientific) and by gel electrophoresis.

### 2.5. Specificity and sensitivity of colorimetric LAMP and the confirmation of LAMP amplicons by restriction enzyme digestion and sequencing

The sensitivity of the assay was confirmed using 10-fold dilutions of cultured bacterial cells (*S. agalactiae*; $1.23 \times 10^3$ to 1.23CFU; *S. iniae*; $1.14 \times 10^3$ to 1.14CFU) as templates with the optimized conditions for colorimetric LAMP as described above. Specificity of the assay was tested using 100 ng of DNA extracted from *S. agalctiae* and *S. iniae* (positive control) and 100 ng of DNA extracted from non-target pathogens including *Straphylococcus epidermidis*

ATCC 12228, *Aeromonas salmonicida* ATCC 14174, *A. hydrophila* ATCC 35654, *Lactococcus garviae* ATCC 49156, *Edwardsiella tarda* ATCC 15947, *Vibrio anguillarum* ATCC 19264, *Shewanella* sp. Laboratory strain, and *Psuedomonas aeruginosa* ATCC 27853. Furthermore, LAMP amplicons were confirmed by restriction enzyme. Positive resulting LAMP amplicons from both *S. aglactiae* and *S. iniae* were digested with *Xmn*I and *Bsr*I restriction enzyme (New England BioLabs) at 37° C and 65° C, respectively, for 1 h followed by 3% agarose gel electrophoresis. The enzyme digested fragments of 165 bp and 152 bp were predicted as the target, which was based upon the predicted amplicon sequence of *S. aglactiae* and *S. iniae*, respectively from *sod*A gene, determined using the Bioedit program (Hall, 1999). The positive samples from colorimetric LAMP assay were subjected to sequencing analysis to confirm the target sequences. The amplified products were cut with *Xmn*I and *Bsr*I restriction enzyme for *S. aglactiae* and *S. iniae,* respectively, and then the digested fragments were cloned into pJET vector (Thermo Fisher Scientific). Plasmid DNA was subsequently purified using an extraction kit (Favorgen) and 50 ng$\mu$l$^{-1}$ of plasmid DNA was then sequenced directly using SP6 universal primer (Macrogen). The *sod*A gene sequences were aligned with published sequences available on the GenBank database (NCBI) using ClustalW.

## 3. Results and Discussion

### 3.1. Colorimetric LAMP assay

In this study, the colorimetric LAMP assay which was established and optimized for the visual detection of streptococcal pathogens in cultured tilapia, with the pre-addition of hydroxynapthol blue (HNB) consistently resulted in the clear discrimination of the positive and negative control samples. Due to the pre-addition of metal dye indicator (HNB), the colour of LAMP mixture was visualized as purple. During positive LAMP reactions, pyrophosphate ion released from the deoxyribonucleotide triphosphate (dNTPs) reacts with magnesium ions to produce a by-product of magnesium pyrophosphate (Goto et al., 2009; Tomita et al., 2008) which shifts the pH to 8, resulting in a colour change from purple to blue. While for the negative reaction the samples remain purple, as there is no production of magnesium pyrophosphate (Figs. 1 A and 2 A ). In our study, the sensitivity of colorimetric LAMP assays for *S. agalactiae* and *S. iniae* was found to be 1.23 and 1.14 CFU, respectively without any cross-amplification when detected along with other fish bacterial pathogens. These results are identical to that of LAMP followed by gel electrophoresis (Fig. 1B and 2B) and by UV-visible analysis (Fig. 1C and 2C). Digestion of *S. aglactiae* and *S. iniae* LAMP-positive amplicons with the restriction enzyme *Xmn*I and *Bsr*I followed by agarose gel electrophoresis yielded the expected fragments of 165 bp and 152 bp respectively. There was also a simultaneous loss of the typical ladder-like pattern of the LAMP amplicons on the gels (Fig. 3A and 3B), confirming that the LAMP amplicons arose from the targeted *sod*A region of *S. aglactiae* and *S. iniae*. When those positive amplicons were subjected to *sod*A gene sequencing and aligned with sequences of *Streptococcus* spp. entries in the GenBank database, they found 100% sequence match with *S. aglactiae* and *S. iniae*as reported previously (data not shown) (Suebsing et al., 2013).

### 3.2. Tissue tropism of Streptococcus in red tilapia

*S. agalactiae* and *S.* iniae were widely present in many of the organs (blood, gill, liver, spleen, kidney, brain and reproductive organs) tested in the 20 broodstock fish (Table 3). All 20 broodstock fish had at least one organ where *S. iniae* was detected and 19 fish with *S. agalactiae*. Among all the tissues from red tilapia broodstock tested for pathogens, blood samples had the lowest presence of *S. agalactiae* (10%), whilst the presence of *S. iniae* (75%) was the highest. Our previous observations of pathogen detection from the blood tissue of red tilipia of adult fish from the same source found that the presence of *S. agalactiae* was equally high as that of *S. iniae* (Pradeep, personal observation). The reason for the variability in the detection of the presence of the pathogens could be related to seasonal variation because activity of the pathogens in tilapia is highly correlated with temperature (Amal et al., 2010; Yuasa et al., 1999). It has been suggested that *Streptococcus* sp. can withstand the bactericidal activity of blood macrophages and multiply in the blood of tilapia (Eldar et al., 1994; Evans et al., 2002; Hernandez et al., 2009; Jantrakajorn et al., 2014; Zlotkin et al., 2003; Zamri-Saad et al., 2010). Many streptococcal species can prevent phagocytosis and opsonization through surface antigens or an extracellular polysaccharide capsule (Fuller et al., 2001; Lindahl et al., 2005; Locke et al., 2007). Failure of initial phagocytosis and killing of the bacteria by the host immune response would thus enhance their survival and systemic proliferation. As macrophages may also act as a carrier for the pathogens, their dissemination to other organs and tissues would potentially be facilitated (Bowater et al., 2012; Evans et al., 2001; Nguyen et al., 2001; Zlotkin et al., 2003) and this may help to explain the widespread presence of the pathogens among the examined organs of the tilapia broodstock.

*Streptococcus* spp. are considered to be neurotropic in tilapia due to the symptoms generated from the central nervous system, such as aberrant fish behavior and swimming patterns (Evans et al., 2000). Lymphocytic infiltration of brain is usually reported in clinical streptococcosis of tilapia (Chang and Plumb, 1996; Suanyuk et al., 2010) and it is one of the most infected organs by *Streptococcus* spp. (Klesius et al., 2008). In the current study, the brain tissue from red tilapia broodstock was confirmed to contain *S. agalactiae* (55%), *S. iniae* (35%) and 25% of individuals hosted both species. This suggest that these pathogens may use immunologically privileged sites, such as the brain, as the immune function in these organs is weak or reduced (Simpson, 2006). Histological findings from naturally infected moribund tilapias have shown severe, extensive meningoencephalitis and uveitis associated with a large number of intracellular cocci presumably related to infection with species of *Streptococcus* (Perera et al., 1998).

Among the 20 broodstock fish the highest incidence of both S*treptococcus* species in the solid tissues was in the liver [*S. agalactiae* (75%) and *S. iniae* (50%)] and spleen [*S. agalactiae* (75%) and *S. iniae* (60%)]. The presence of the pathogens was also commonly detected in other organs such as the gills [*S. agalactiae* (70%) and *S. iniae* (30%)] and kidney [*S. agalactiae* (65%) and *S. iniae* (35%)]. As both bacteria species were detected from some of the organs that are involved in the defence of fish from pathogens, such as the spleen and kidney, it is possible that the high mortality rate as reported for the early developmental phases in farmed tilapia due to streptococcosis could be due to their compromised defence mechanisms. Experimental infection of *S. agalactiae* in red tilapia has revealed that this pathogen can pass though the gastrointestinal epithelium and can accumulate and localize in the spleen as the preliminary organ of infection (Iregui et al., 2015). Although blood could be used as a non-lethal and easy source of pathogen detection our results indicate doing so would be unreliable, as the pathogens are localized more in organs than in blood. For example, the false negative detection rate in the current study using samples of blood compared to all other organs was 90% for *S. agalactiae* and 25% for *S. iniae*. Therefore, liver and spleen tissues would be more appropriate organs for disease diagnostic applications in red tilapia given their more consistent presence in these tissues. Other experimental studies on the pathogenicity of *Streptococcus* spp. in tilapia have recommended that brain, eye and kidney tissues as the main target organs (Abuseliana et al., 2011; Rodkhum et al., 2011).

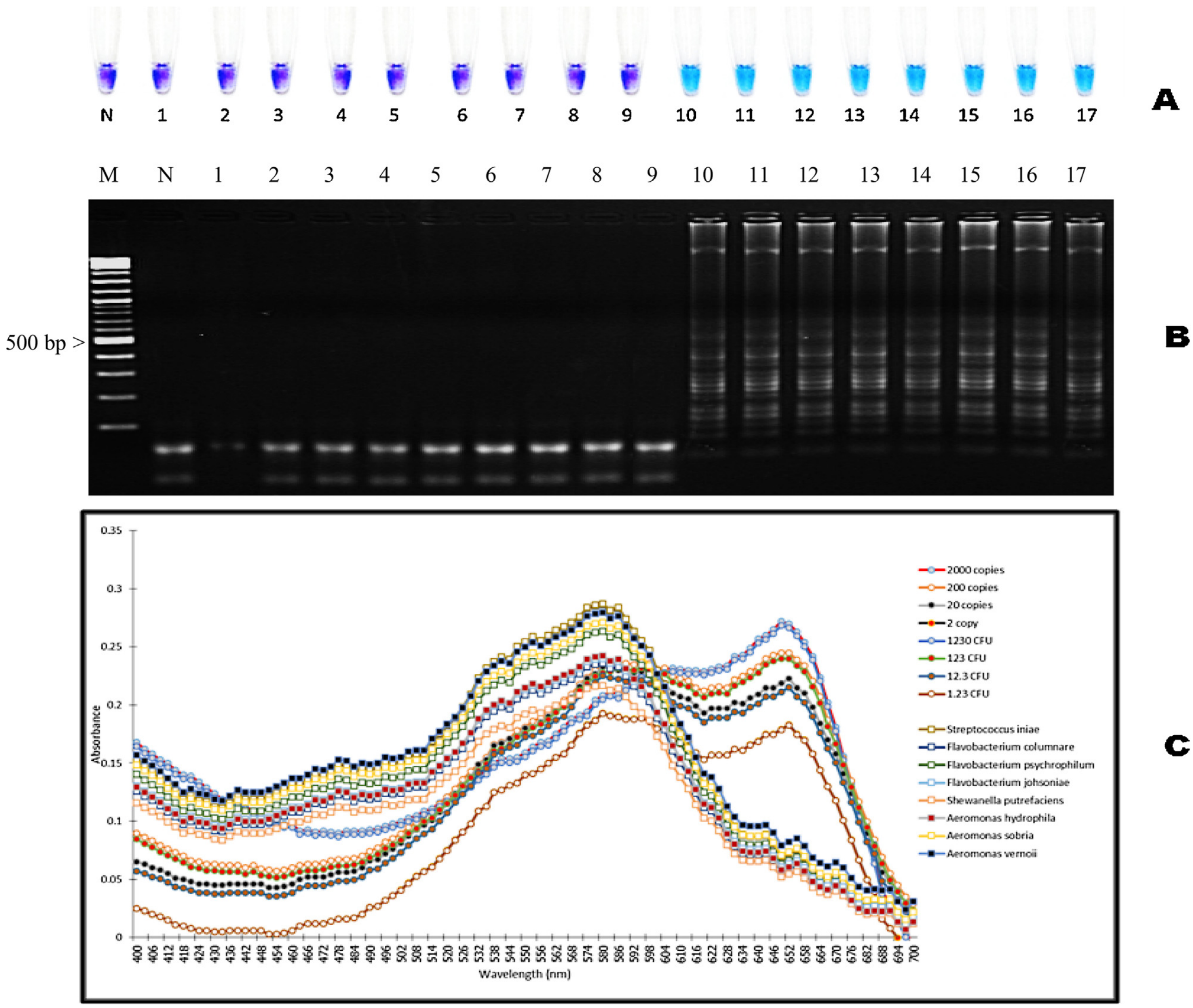


**Fig. 1.** The sensitivity and specificity of colorimetric LAMP for detection of *S. agalactiae* DMST 17129 with; (A) pre-adding hydroxynapthol blue. LAMP followed by; (B) gel electrophoresis of *S. agalacite* and; (C) UV–visible analysis (blue color absorb at OD 650 and purple color absorb at OD 580) corresponding to individual tubes in Fig. 1A. Lane M: DNA marker, lanes 1–9: *S. iniae* ATCC 29178, *S. epidermidis* ATCC 12228, *Aeromonas salmonicida* ATCC 14174, *A. hydrophila* ATCC 35654, *Lactococcus garviae* ATCC 49156, *Edwardsiella tarda* ATCC 15947, *Vibrio anguillarum* ATCC 19264, *Shewanella* sp. Laboratory strain, and *Psuedomonas aeruginosa* ATCC 27853, lanes 10–13: plasmid DNA inserting *sodA* of *S. agalactiae* at $2 \times 10^3$ to 2 copies and lanes 14–17: *S. agalactiae* at $1.23 \times 10^3$–1.23CFU $ml^{-1}$.

**Table 3**
The presence (+) or absence (−) of *S. agalactiae* (SA) and *S. iniae* (SI) in the organs of paired broodstock (i.e., $n$ = 10♀ and 10♂) of red tilapia identified using LAMP assay.

| Sample | ♂-Blood | | ♂-Gill | | ♂-Liver | | ♂- Spleen | | ♂-Kidney | | ♂-Brain | | ♀-Blood | | ♀-Gill | | ♀-Liver | | ♀- Spleen | | ♀-Kidney | | ♀-Brain | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pathogen | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI |
| Family 1 | − | − | − | − | − | − | − | + | − | + | − | − | − | − | − | − | + | + | + | − | − | − | + | + |
| Family 2 | − | + | + | + | + | + | + | + | + | + | − | − | + | + | − | + | + | + | − | + | + | + | + | − |
| Family3 | − | + | + | + | + | + | + | − | + | − | + | + | − | + | + | − | + | − | + | + | + | − | − | + |
| Family 4 | − | + | + | + | + | − | − | − | + | − | − | − | − | + | + | + | + | − | + | − | + | − | + | − |
| Family 5 | − | + | + | + | + | + | + | + | − | − | − | + | − | + | − | − | + | + | + | + | + | + | + | + |
| Family 6 | − | + | + | − | − | − | − | + | − | − | − | − | − | + | + | − | + | − | + | + | + | + | + | + |
| Family 7 | − | + | + | − | − | − | + | + | + | − | + | − | − | + | + | − | + | − | + | − | + | − | + | − |
| Family 8 | − | - | − | − | + | − | − | − | + | + | − | − | − | + | + | − | + | + | + | + | + | − | + | − |
| Family 9 | − | + | − | − | − | + | + | + | − | + | − | + | − | − | + | − | + | + | + | + | − | − | + | − |
| Family 10 | + | − | + | − | − | + | + | − | + | − | − | − | − | + | + | − | + | − | + | − | − | − | + | − |
| Total | 1/10 | 7/10 | 7/10 | 4/10 | 5/10 | 5/10 | 6/10 | 6/10 | 6/10 | 4/10 | 2/10 | 3/10 | 1/10 | 8/10 | 7/10 | 2/10 | 10/10 | 5/10 | 9/10 | 6/10 | 7/10 | 3/10 | 9/10 | 4/10 |

## 3.3. *Vertical transmission*

No mortality or any clinical signs of disease was observed in the broodstock fish during the entire experiment, despite all 20 being infected with *S. iniae* and 19 with *S. agalactiae*, so that all 10 pairs had the potential to pass on both pathogens via gametes (Table 4). All resulting milt, unfertilized and fertilized egg and one-day-old larval samples were detected as negative for *S. agalactiae* through LAMP analysis, despite being present in the reproductive organs of 8 out of 10 broodstock pairs. However, some 10 and 30-day-

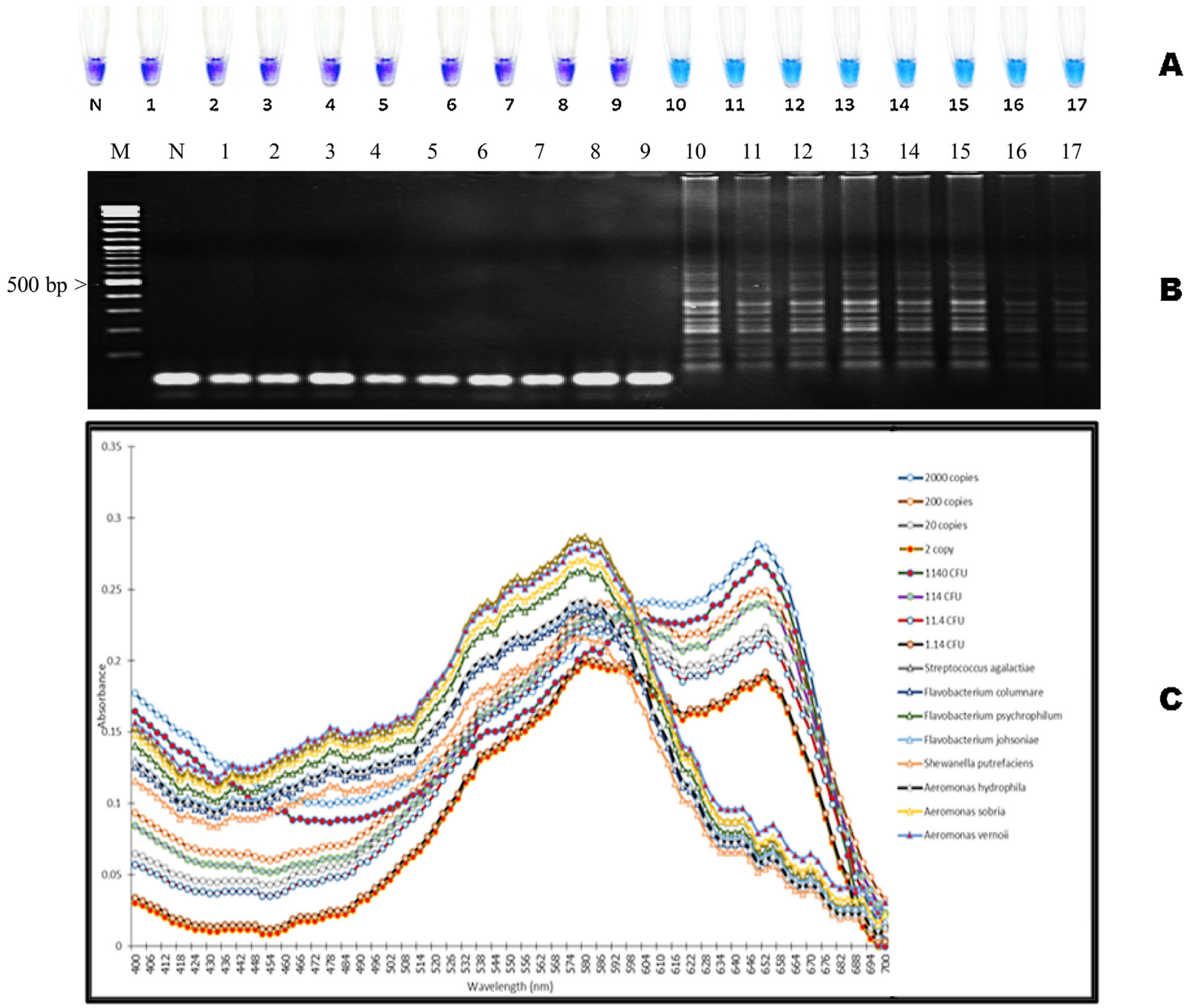


**Fig. 2.** The sensitivity and specificity of colorimetric LAMP for detection of *S. iniae* ATCC 29178 with; (A) pre-adding hydroxynapthol blue. LAMP followed by; (B) gel electrophoresis of *S. iniae* and; (C) UV–visible spectrometer analysis (blue color absorb at OD 650 and purple color absorb at OD 580) corresponding to individual tubes in Fig. 2A Lane M: DNA marker, lanes 1–9: *S. agalactiae* DMST 17129, *S. epidermidis* ATCC 12228, *A. salmonicida* ATCC 14174, *A. hydrophila* ATCC 35654, *Lactococcus garviae* ATCC 49156, *Edwardsiella tarda* ATCC 15947, *Vibrio anguillarum* ATCC 19264, *Shewanella* sp. Laboratory strain, and *Psuedomonas aeruginosa* ATCC 27853, lanes 10–13: plasmid DNA inserting *sodA* of *S. iniae* at $2 \times 10^3$ to 2 copies and lanes 14–17: *S. iniae* at $1.14 \times 10^3$ to 1.14 CFU ml$^{-1}$.

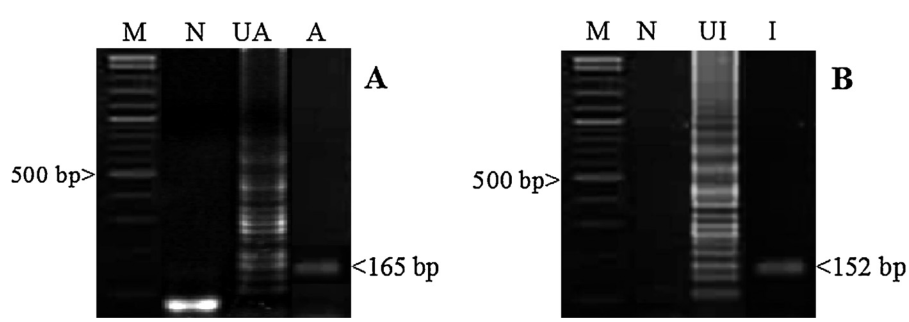


**Fig. 3.** Confirmation of colorimetric LAMP amplicons by restriction enzyme analysis of; (A) *S. agalactiae* and; (B) *S. iniae*. M: 2 log DNA marker, N: negative control, UA: uncut *S. agalactiae* LAMP amplicon, UI: uncut *S. iniae* LAMP amplicon, A: *Xmn*I enzyme digested fragment of 165 bp, I: *Bsr*I enzyme digested fragment of 152 bp.

old larval groups tested positive for the presence of this pathogen (both stages at 4/10). In comparison, *S. iniae* was common in milt (9/10) and unfertilized eggs (7/10) and continued to be detected in later stages such that it was present in all 10 family groups of 30-day-old larvae (10/10) (Table 4). The lack of a positive result for both pathogens from earlier stages of development for probably relate to their presence being below detectable levels using the LAMP method, but with time they have multiplied within the

**Table 4**
The presence (+) or absence (−) of *Streptococcus agalactiae* (SA) *and S.iniae* (SI) in the samples of red tilapia by colorimetric LAMP assay.

| Family | Sexual organs Testes | | Gonad | | Sample source Milt | | Unfertilized egg | | Fertilized egg | | One-day-old larvae | | Ten-day-old larvae | | Thirty-day-larvae | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI | SA | SI |
| Family 1 | – | + | – | + | – | + | – | + | – | – | – | + | – | + | – | + |
| Family 2 | – | + | + | + | – | + | – | + | – | – | – | + | – | + | – | + |
| Family 3 | + | + | + | + | – | + | – | + | – | + | – | + | + | + | – | + |
| Family 4 | – | + | – | + | – | + | – | + | – | + | – | + | – | + | – | + |
| Family 5 | – | + | + | + | – | + | – | – | – | + | – | + | – | + | + | + |
| Family 6 | – | + | + | + | – | + | – | + | – | + | – | + | + | + | + | + |
| Family 7 | + | + | + | + | – | + | – | + | – | + | – | + | + | + | – | + |
| Family 8 | – | – | + | + | – | – | – | + | – | + | – | – | – | + | – | + |
| Family 9 | + | + | + | + | – | + | – | – | – | – | – | + | – | – | + | + |
| Family 10 | – | – | + | + | – | + | – | – | – | – | – | + | + | – | + | + |
| Total | 3/10 | 8/10 | 8/10 | 10/10 | 0/10 | 9/10 | 0/10 | 7/10 | 0/10 | 6/10 | 0/10 | 9/10 | 4/10 | 8/10 | 4/10 | 10/10 |

rapidly developing larvae to sufficient concentration for detection. *S. agalactiae* was not detected in any samples of milt, unfertilized or fertilized eggs, or one-day-old larvae.

Analysis of reproductive organs from the broostock confirmed that both *S. agalactiae* and *S. iniae* can be found in both testes and ovaries with relatively high incidence; *S. agalactiae* [testes (3/10) and ovary (8/10)] and *S. iniae* [testes (8/10) and ovary (10/10)]. For *S. agalactiae* there were three crosses (families 5, 6, 10) when the gametes or offspring were positive for the pathogen whereas the testes of the paternal broodstock tested negative, while the ovaries of the maternal broodstock tested positive. The same was true for *S. iniae* for two crosses (families 8, 10). This suggests vertical transmission of both pathogens via ovaries to eggs is common. Unfortunately, no crosses were made where testes were positive for pathogens whereas the corresponding ovaries were not, making it difficult to infer transmission via testes to milt.

The results from the present study has thus revealed that in contrary to the widely held belief that streptococcosis is mostly transmitted through horizontal means, especially through movements of infected fish (Amal et al., 2013b; Nguyen et al., 2002), cannibalism of the dead or moribund fish (Plumb, 1997, 1999; Shoemaker et al., 2000) and through bacterial contamination of water in culture systems (Agnew and Barnes, 2007; Amal et al., 2013a; Bowater et al., 2012; Evans et al., 2002; Hossain et al., 2014; Kim et al., 2007; Mian et al., 2009; Nguyen et al., 2002; Robinson and Meyer, 1966; Rasheed and Plumb, 1984; Shoemaker et al., 2000; Xu et al., 2007)], the chances of the pathogens being transmitted through vertical transmission is also likely in red tilapia.

One of the practical means to predict the vertical transmission of pathogens from broodstock to their progeny is by experimentally infecting pathogen-free broodstock and crossing them to produce progeny for the subsequent evaluation of the presence of the pathogen (Bootland et al., 1991; Dorson and Torchy, 1985). However, in our study we have used randomly selected broodstock from a farm with a previous streptococcosis outbreak. Our assumption was that the sample of broodstock would be asymptomatic carriers of the pathogen with the potential to pass the pathogens onto their progeny. Two streptococcosis pathogens, *S. agalactiae* (♂ = 3/10 and ♀ = 8/10) and *S. iniae* (♂ = 8/10 and ♀ = 10/10) were present in the reproductive organs of the broodstock and were subsequently detected in various developmental stages of most of their progeny. This was consistent with our previous study where the presence of both *S. iniae* and *S. agalactiae* were found in the reproductive organs (testis and ovary), as well as in fertilized eggs and fingerlings derived from pooled fertilizations (Suebsing et al., 2013). Recently, histopathological observations of streptococcal bacteria in ovarian tissues of tilapia was reported (Jantrakajorn et al., 2014), which clearly revealed that reproductive organs can be a favorable site for *Streptococcus* spp. in tilapia, with multiple pathogens present in the same organ of the host. In the current study, streptococcal pathogens could be detected even at early development stages of tilapia, contrary to earlier reports suggesting they could only be detected in fish greater than 20 g in size (Hernandez et al., 2009; Jimenez et al., 2011).

### 3.4. Prevalence of bacteria in water sample

Streptococcal species can survive in water and sediments of aquaculture systems (Nguyen et al., 2002). In the present study both *S. agalactiae* and *S. iniae* were found in water sampling. *Streptococcus agalactiae* was confirmed in a small number of all the water sources (BT = 10%, HC = 10%, ST-10Day = 10%, ST-30Day = 10%), while the presence of *S. iniae* was more commonly detected [(BT = 70%; HC = 70%, ST-10Day = 30%, ST-30Day = 80%). The water temperature in the indoor culture facility from which BT (25 ± 1 °C) and HC (27 ± 1 °C) samples were taken was mostly below 27 °C which could be expected to have favoured the proliferation of *S. iniae* compared to *S. agalactiae* because outbreaks of *S. agalactiae* in aquaculture operations are most common at water temperatures above 27 °C (Mian et al., 2009), although they can sometimes occur at lower temperatures (Agnew and Barnes, 2007; Mian et al., 2009; Yuasa et al., 1999).

## 4. Conclusions

The results of the present study confirm that both *S. agalactiae* and *S. iniae* can be vertically transmitted in cultured red tilapia. Furthermore, the pathogens can be passed on by non-symptomatic broodstock infected with the pathogens. The high proportion of families where vertical transmission occurred (60% in *S. agalactiae*, 100% in *S. iniae*) suggests that pathogens cannot easily be eliminated from aquaculture stock using an intergenerational break, which is a common method employed in other cultured species. Therefore, the elimination of these pathogens from aquaculture stock of red tilapia will rely upon the careful application of chemotherapy both to broodstock and offspring, and subsequent good management and biosecurity practice to prevent reinfection (Amal and Zamri-Saad, 2011). The development of specific pathogen-free tilapia, especially as a founding broodstock population, has the potential to be a valuable strategy to control assist in controlling streptococcosis, which is an aggressive disease constantly threatening the sustainability of the global tilapia industry.

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